

Gate E8 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS with findings**. Evidence directory: 20_EVIDENCE/E8/.

E8 — Internal evidence and packaging (preliminary, after independent first pass)

Independent first-pass records saved before any author verdict/derivation was read: E1_RECORD.md, E3_STATIC_RECORD.md, E7_FIRST_PASS.md + E7_LEAD_VERIFICATION.md, E2_RECORD.md, E5_RESULTS.json/E5_RECORD.md, E6_RECORD.md (timestamps in the files). Only then were read: author build evidence (r2, resume, recovery/provenance/validation JSON), internal-audit final report, FINDINGS.md, MATHEMATICAL_REVIEW.md §A.2, B01 Certo run. Access logged.

A. Author two-segment reconstruction (script `e8_author_build_check.py`, output `E8_AUTHOR_BUILD_CHECK.json`)

- Segment 1 (`full_rebuild_v12_20260929_r2`): 422 PASS rows, status file preserved as `RUNNING 423/607 E34.SampleBound` (interruption evidence preserved). All 422 exits 0, logs present, no sorry/error; sources equal the manifest.
- Segment 2 (`...resume`): 607 rows = 421 UP-TO-DATE + 186 PASS; union = 607 distinct, no overlap; `BUILD.exit 0`; no failed/blocked/not-run; sources unchanged.
- Every reused module has a segment-1 PASS with identical source hash and library; run metadata (Lean binary/version, options, threads, dependency pins, `LEAN_PATH`, targets) identical across segments; `RECOVERY_CHECK` olean hashes equal segment-1 `RESULTS` olean hashes for all 422; `A4S1.IndependAllAudit` (a target) passed in segment 1 and was re-executed in segment 2 (the "one re-executed completion").
- All 19 targets PASS with exit 0 in segment 2; no `sorryAx/sorry`; no non-standard axiom.
- Limitation: reuse validity also depends on the author runner's dependency-olean trace files, which are in the author work directory and not in the evidence set; the auditor could verify record consistency, not re-hash the reused objects.
- Axiom-record count reconciled:** the author's `regex (?::depends on axioms:|axioms)\s*\[...\]` yields exactly 461 over the 19 target logs; of these 314 are genuine `#print axioms` outputs and 147 are axiom lists printed by the custom audit elaborators (ResearchAudit 75, E34 extra, etc.). The internal report cites "313 nominal prints over 307 distinct declarations" (1-record difference from the auditor's 314, OBSERVATION E8-O1). The manuscript correctly says the 461 records are not distinct theorems.
- This is author-build evidence. It does not replace E4.

B. Internal audit (`PASS_INTERNAL_AUTHOR_SIDE`) — challenges

#	Internal position	Auditor challenge	Effect
B1	All seven A.2 items <code>ACCEPTABLE_SUMMARY</code> "for the paper accompanied by its sources"	The mandate asks for adequacy for a human referee, not with the Lean sources. Internal item 5 itself says normalization is "exposition by contracts with named implementations, not a proof in prose". Internal item 3 (fibres) did not detect that per-edge feasibility needs the two-sided deviation cleanup, absent from the prose. Auditor: A2-3, A2-4, A2-5, A2-6 <code>REQUIRES_EXPANSION</code> .	Disagreement preserved; auditor verdicts stand (E2).
B2	G5/G6 bilingual & visual PASS	Missed E6-01 ("holgura" for margin in ES, contradicting its own 1.159 terminology), E6-03 (Spanish subscripts in EN math), E6-04 (A.1 "literal" Lean excerpt rendered <code>{\mathbb{V} : \mathbb{V}}</code>).	MINOR/MODERATE findings added.
B3	I-05: '§3.2' in A.2 is an erratum kept outside the sealed manuscript	Confirmed: EN A.2 row "§3.2 and C.2" has no target section. Stays in the audited manuscript.	MINOR finding (F-A2REF).
B4	Certo 0.20.1 certificate for K3v13	Audited by its actual specification: universe = the 12 edges of K3v13; 6 parts; each part is the full edge set of a clique; exact cover (auditor's own checker, no Certo import). The certificate proves feasibility (a 6-part partition) only; Certo itself warns it is not optimality. Optimality $6 = kh - C(k,2)$ comes from the weight argument (6.1). The tamper run lists <code>cliques/exact</code> as "excused" mutations, so the clique property was not tamper-tested by Certo; the auditor's checker covers it.	E5 T25 updated: Certo item AUDITED (feasibility-only, correct).
B5	"No other build was launched"	Internal G4 relied on recorded author builds; E4 of this run is the first external isolated build.	—

C. Packaging

- Internal package ZIP 488 members / 487 manifest entries (`PACKAGE_VERIFICATION`); hash equals the binding (E0).

- Author evidence ZIP 1239 = 1238 + manifest; all hashes verified (E0).

Status: **preliminary E8 PASS on evidence consistency; disagreements B1–B3 recorded as findings.** Final E8 after E4 (compare own build outputs with author records).

Final E8 (after E4)

- The auditor's isolated build reproduces all 607 author olean hashes and all 19 target logs byte for byte (apart from the command line), which corroborates the two-segment reuse record (X-21 resolved).
- The author runner's per-module sorry detection uses the pattern `declaration uses 'sorry'`, which Lean 4.28 does not print (it uses backticks). The internal/author validation therefore relied on it ineffectively, though no sorry exists in any log (X-28, MINOR).
- The internal audit did not notice X-27 (cp classification cone on the historical Alon–Shapira route).

E8 verdict: PASS with findings (disagreements B1–B3, X-27, X-28 recorded).

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